

Additions for review — Document 3, page 8, section 4

"How we approach investment decisions", from the extract of the board strategy day. Each section below is drafted separately so it can be reviewed and placed where it fits best. The grey line under each heading states where it was drafted to sit.

1. Liquidity and covenant impact

Drafted to sit: Final Decision, as the second bullet on page 8. Nothing in the research currently covers it.

A positive NPV does not mean Axion can fund the project. Page 8 asks this as a separate question because the pipeline is R3 to R5 billion over five years and borrowings have risen every year.

Measure	FY2024	FY2025	FY2026	What it shows
Interest cover (EBITDA / finance costs)	3.31	3.12	2.98	Falling every year despite EBITDA growth
Interest cover (operating profit / finance costs)	2.04	1.95	1.87	Below 2 times on the more conservative measure lenders use
Borrowings / EBITDA	4.23	4.36	4.35	At the level at which most lenders set the covenant
Borrowings / equity	0.78	0.85	0.90	Rising
Gearing, borrowings / (borrowings + equity)	43.8%	45.9%	47.3%	Rising
Capital expenditure / EBITDA	—	2.0	1.8	Capital spend is running at roughly twice operating cash generation

Calculated from Document 5, section 3. What to test on the day:

- Headroom: recompute each ratio including the new debt and the new EBITDA, in the worst year of the forecast, not on an average.
- Debt service cover: project cash flow available for debt service divided by principal plus interest, tested each year rather than once.
- Timing mismatch: debt is drawn and interest starts during construction, while the electricity benefit only starts at commissioning.
- Consequences of a breach: cure rights, an equity cure, and whether a breach on a project facility cross-defaults to the existing group facilities.
- Opportunity cost: every rand of debt capacity used on generation is not available for data centre capacity expansion, which is the GROW priority.

Applied to Axion: the group is already at about 4.35 times borrowings to EBITDA and below 2 times operating profit interest cover, before any renewable energy spend. Debt-funding a plant of this size pushes both measures the wrong way during construction, when there is still no electricity benefit. That is the strongest financial argument for the PPA route, or for a structure in which the debt genuinely sits outside the group.

2. Risk allocation and safeguards

Drafted to sit: Not a new table. These are the three new columns to paste onto the existing risk tables under "Principal assumptions and risk". The rows follow the same order, so they line up one for one.

The research identifies the risk and the impact. Page 8 asks the next two questions: who carries it, and what safeguard reduces the residual exposure.

Existing table — PPA

Risk already identified	Who carries it	Safeguard to negotiate	Residual exposure to Axion
Future customer requirements and power-density profiles differ from assumptions, particularly where AI-related demand grows faster or slower than expected	Axion. It sets the contracted volume	Volume flexibility band, a right to on-sell or wheel surplus, a shorter initial term with extension options, and volume step-up rights as load ramps	Take-or-pay cost on volume not used, or emergency procurement at spot if demand outruns the contract
Curtailment, wheeling losses, grid constraints or supplier underperformance reduce delivered electricity	Shared. Curtailment and network losses usually sit with Axion unless negotiated	Deemed generation clause, an availability warranty, a minimum yield undertaking, and explicit allocation of loss and curtailment risk in the wheeling agreement	Network charges may still be payable on power not received; replacement power bought from the grid
Electricity tariffs, wheeling charges or regulatory costs	Shared. Tariff sits with the	Cap on escalation, a fixed tariff for an agreed period, and a change in	Network and regulatory cost increases largely

increase above forecast	producer; network and regulatory cost usually with Axion	law clause allocating regulatory cost	remain with Axion
Contractual restrictions prevent monetising excess electricity	Axion	An express right to sell, wheel, bank or receive credit for surplus, and confirmation of how surplus is settled in each interval	Where surplus cannot be monetised, the take-or-pay cost is a pure loss
Supplier financial distress or operational failures disrupt supply (counterparty risk)	The producer	Parent company guarantee, step-in rights, security over the plant, and the right to terminate and replace	Cost and delay of replacing supply, and reliance on grid electricity in the interim

Existing table — Investment in plant

Risk already identified	Who carries it	Safeguard to negotiate	Residual exposure to Axion
Actual customer demand, utilisation trends or customer mix differ from forecast, particularly if AI-related demand develops differently from expectations	Axion	Phase the build so capacity is added as load ramps, and secure a right to sell surplus output	Stranded generation capacity, or insufficient capacity requiring grid purchases
Higher operating and maintenance costs or earlier-than-expected equipment replacement	Axion, unless contracted out	A long-term operating and maintenance contract with performance standards and price certainty, plus equipment warranties	Cost above the contracted scope, and replacement capital earlier than modelled
Weather, equipment failure, curtailment, transmission loss or operating inefficiency reduce generation	Axion as owner	Performance guarantee from the contractor, an availability warranty, and an independently verified yield study	Resource variability is not contractible; grid purchases make up the shortfall
Delays, contractor issues or cost overruns during construction	The contractor	Fixed-price turnkey contract, delay damages, a defined commercial operation date with remedies, retention and performance bonds	Damages rarely cover a full loss of benefit, and the benefit is deferred
Regulatory changes, approval delays or grid access constraints	Axion	Confirm grid capacity allocation and the wheeling agreement before financial close; make the contract conditional on approvals	Approvals sit on the critical path to the first year of benefit in the model

Existing table — AI-related risk (both alternatives)

Risk already identified	Who carries it	Safeguard to negotiate	Residual exposure to Axion
AI customers require significantly higher rack densities and power consumption than anticipated	Axion	Contract for capacity in tranches with pre-agreed step-up pricing, and align the electricity procurement forecast to the same demand model used for revenue	Emergency procurement at spot, and accelerated expansion capital

3. Required approvals

Drafted to sit: Final Decision, as the fourth bullet on page 8.

Approval	Why it is needed	What to confirm
Board approval	The delegation of authority framework sets the value at which board approval is required	Whether the project exceeds the threshold, and whether the board has approved the funding as well as the project itself
Shareholder approval	The memorandum of incorporation and any shareholders agreement. Axion has five shareholders and none holds control	Whether capital expenditure of this size, or entering a joint venture, is a reserved matter requiring shareholder consent
Lender consent	Existing facility agreements commonly restrict additional borrowing, the granting of security and disposals	Whether the new funding or security breaches a negative pledge and needs a waiver before signature
Companies Act section 45	Financial assistance to a related or inter-related company. A loan or guarantee to a joint venture or project company falls within this	A special resolution passed within the previous two years, the board applying the solvency and liquidity test, and written notice to shareholders and any trade union. Non-compliance makes the decision void and exposes directors personally under section 77(3)(e)(iv)
NERSA registration	Generation above the registration	The timeline, because regulatory approval sits on the critical

or licensing	threshold, and any wheeling arrangement	path to the commercial operation date and therefore to the first year of benefit in the model
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4. Funding should reflect the project

Drafted to sit: Its own section, after Final Decision. Page 8 gives this its own band and the research does not cover it.

Page 8 states that the funding structure appropriate for Axion's existing operations may not be appropriate for every new project. The three headings below are taken directly from page 8.

Project cash flows — predictability, duration, contractual support

Feature	Additional PPAs	Direct investment	Effect on debt capacity
Predictability	No new revenue stream. The effect is on Axion's operating cost	Generation varies with the resource, and the value of the saving varies with the grid tariff	A less predictable benefit supports less debt
Duration	Typically 3 to 5 years	25 to 30 years	Longer duration supports longer tenor debt
Contractual support	The PPA is itself the contract, but it is a purchase obligation, not a receivable	Only where output is sold under an offtake contract. A plant built for Axion's own consumption produces a cost saving, not contracted revenue	This is the decisive point. A self-consumption plant has no contracted revenue stream to pledge, so it supports materially less project debt than an independent producer selling under a long-term PPA

Target capital structure — project-specific mix

- The mix should reflect the risk of the project, not Axion's current group gearing of about 47%.
- A contracted, long-duration generation project can usually carry more debt than a data centre build. A self-consumption plant with no contracted offtake carries less.
- Whatever mix is chosen must then be used to regear the beta in section 5. Deriving the discount rate from one capital structure and funding the project at another is internally inconsistent, and is an easy point to be caught on.

Security and support — and why project-level funding does not transfer risk

Page 8 warns that project-level funding does not necessarily transfer risk away from Axion. Test that warning against what is actually signed.

- Sponsor support brings the risk back: completion guarantees, cost overrun undertakings and equity contribution agreements are all Axion obligations.
- Security typically includes a pledge of the shares in the project company, cession of the project contracts and a debt service reserve account funded by Axion.
- Project covenants trap cash: a distribution lock-up until a debt service cover ratio is met withholds the very cash flows Axion was relying on.
- Accounting follows substance. If Axion controls the project company it consolidates under IFRS 10 and the debt appears on the group statement of financial position. Joint control gives equity accounting under IAS 28, which keeps the debt off the face of the statements but not out of the group's risk. Axion already equity accounts its 40% interest in Metroloop, so the precedent exists.
- Operational recourse remains regardless of structure. If the plant fails, Axion must still power the data centres and meet the 99.999% availability commitment.

5. Project-specific cost of capital

Drafted to sit: Its own section. Page 8 gives the three-beta flow its own band; the research currently mentions it in one bullet.

The purpose of the three steps on page 8 is to obtain a discount rate reflecting the risk of the renewable energy project, not the risk of Axion's existing data centre operations. Axion is unlisted and has no observable beta of its own, and in any event a generation project is a different business.

Beta	What it measures	Where it comes from
Comparable company equity beta (also called the geared or levered beta)	Operating risk and financial risk of a listed comparable	Observed from the comparable's share price movements against the market
Underlying beta (also called the asset or ungeared beta)	Operating risk only, with the effect of gearing removed	Calculated by ungearing the comparable's equity beta at the comparable's own capital structure

Debt beta	The systematic risk of the debt itself	Usually assumed to be nil. If the debt is genuinely risky, assuming nil understates the asset beta
Project equity beta (regeared)	Operating risk of the project plus the financial risk of the project's own capital structure	Calculated by regearing the asset beta at the project's target debt to equity mix

The formulas

- Ungear, assuming a debt beta of nil: $\beta_{\text{asset}} = \beta_{\text{equity}} \times E / (E + D(1 - t))$
- Regear: $\beta_{\text{equity}} = \beta_{\text{asset}} \times (1 + D(1 - t) / E)$
- Ungear where a debt beta is given: $\beta_{\text{asset}} = \beta_{\text{equity}} \times E / (E + D(1 - t)) + \beta_{\text{debt}} \times D(1 - t) / (E + D(1 - t))$
- Without tax: $\beta_{\text{asset}} = \beta_{\text{equity}} \times E / (E + D)$
- CAPM: $K_e = R_f + \beta_{\text{equity}} (R_m - R_f)$
- WACC = $K_e \times E / (D + E) + K_d (1 - t) \times D / (D + E)$
- Pre-tax equivalent, as an approximation only: $\text{pre-tax rate} = \text{post-tax WACC} / (1 - t)$

Worked example

Step	Calculation	Result
1. Comparable listed renewable energy producer	Equity beta 0.90, debt to equity 60:40, tax 27%	$\beta_{\text{equity}} = 0.90$
2. Ungear at the comparable's structure	$0.90 / (1 + 1.5 \times 0.73)$	$\beta_{\text{asset}} = 0.430$
3. Regear at the project's target mix of 50:50	$0.430 \times (1 + 1.0 \times 0.73)$	$\beta_{\text{equity}} = 0.743$
4. Cost of equity, risk-free rate 9.5% and market risk premium 6%	$9.5\% + 0.743 \times 6\%$	$K_e = 13.96\%$
5. Cost of debt after tax, pre-tax cost 12%	$12\% \times 0.73$	$K_d = 8.76\%$
6. Weighted average cost of capital	$(0.5 \times 13.96\%) + (0.5 \times 8.76\%)$	$\text{WACC} = 11.36\%$
7. Pre-tax equivalent, for use with pre-tax cash flows	$11.36\% / 0.73$	15.56%

The inputs above are illustrative. Use the figures given on the day.

What to watch

- Use several comparables where they are given: ungear each at its own capital structure, average the asset betas, then regear once at the project mix. A single comparable imports that company's own peculiarities.
- The comparable must match the project, not Axion. For the generation project the comparable is a renewable energy producer, not a data centre operator.
- Regear at the project's target capital structure, not at its opening structure and not at the group's.
- Match the rate to the cash flows: real with real, nominal with nominal, pre-tax with pre-tax. The research uses pre-tax cash flows at initial feasibility.
- The risk-free rate and market risk premium must be South African if the cash flows are in rand.
- Do not add an arbitrary premium on top of a beta-derived rate for risks already in the cash flows. Project-specific risks belong in the cash flows or in sensitivity analysis, not in the discount rate.

6. Break-even — two refinements

Drafted to sit: Appended to the existing "Break-even analysis" section. The variables already listed there are the right ones and need no change. These two points change how the answer is framed.

(a) The break-even that matters is the point of indifference, not an NPV of zero

- Axion is choosing between two ways of buying the same electricity. Both alternatives are largely cost structures, so each can show a negative NPV on its own and one can still be clearly better.
- Setting each alternative's NPV to zero answers "should we do this at all". Setting the NPV of the PPA equal to the NPV of the direct investment answers "which one", which is the question the CFO is actually asking.
- Solve for the variable in question. The PPA tariff at which the two are equal is the maximum tariff Axion should accept: above it, build; below it, contract. The same logic gives a maximum acceptable capital cost per MW for the plant.
- State the answer as a decision rule, not a number. "Contract at any tariff below R1.15 per kWh, otherwise build" is useful to a CFO; "the break-even NPV is nil" is not.

(b) Express both alternatives as a levelised cost per kWh

- Levelised cost = the present value of all costs over the life, divided by the present value of the electricity delivered over that same life, both discounted at the project rate.
- It puts a 3 to 5 year PPA and a 25 year plant on the same basis. A raw NPV comparison does not, because the alternatives have different lives and deliver different volumes.
- For the plant, include capital, operating and maintenance, replacement capital and the tax shield. For the PPA, include the tariff plus wheeling and network charges, less any credit for surplus.
- The comparison then reads directly: if the plant's levelised cost is below the PPA tariff, the plant is cheaper per unit before any qualitative factor is considered.
- The PMT calculation already in the research is the equivalent annual cost technique. Name it as such, and use it to compare alternatives with different lives rather than only to find a break-even income.

7. Use of digital tools and independent validation

Drafted to sit: Its own short section. Page 8 gives this its own box, and the research does not deal with it.

Page 8 states that digital tools support the analysis and calculations but do not replace judgement, and that outputs are reviewed and independently validated before being used. If a model or an AI-generated analysis is given on the day, the task is likely to be to critique it rather than to accept it.

What independent validation of a financial model actually means:

- The reviewer is not the preparer, and the review is documented.
- Recalculate a sample of the formulas and test the logic of the model, not only its outputs.
- Agree the inputs to source: tariffs to the contract, capacity and yield to the technical report, tax rates to the Act, and financial data to the management accounts.
- Check internal consistency: the same demand forecast must drive both the revenue model and the electricity procurement model. If revenue assumes AI-driven growth and the electricity model does not, one of them is wrong.
- Run the sensitivities independently rather than accepting the ones presented, and test the variables that can change the recommendation.
- Confirm version control, and that the version approved is the version presented to the board.
- Confirm that assumptions are documented, sourced and approved by someone with the authority to approve them.

8. Critique of the approach in Document 3

Drafted to sit: Its own section, or material for a task asking you to review and critique the approach. Reviewing someone else's work is the most common task shape in recent papers.

Observation	Why it matters, and what to recommend
What works. The framework moves correctly from economic viability, to funding, to approval, and separates whether the project is worth doing from whether Axion can fund it. Risk is considered at every stage.	Acknowledge this before criticising anything. A critique that finds nothing good is not a critique.
Initial feasibility uses pre-tax cash flows and a pre-tax discount rate.	Internally consistent, but it makes the capital allowance benefit invisible at the screening stage, which is exactly where direct investment differs most from a PPA. A project can screen out pre-tax and be positive after tax. Recommend that tax be brought in at screening where the alternatives have materially different tax profiles.
Project-specific cost of capital sits at initial feasibility, but the beta flow at the foot of page 8 needs the project's capital structure, which is only settled later.	The process is iterative and the pack presents it as linear. Recommend stating the assumed structure used at screening and revisiting the rate once funding is agreed.
Risk is identified at initial feasibility but allocated only at final decision.	Safeguards cost money. Recommend pricing them into the cash flows rather than adding them as conditions after the NPV is calculated.
Nothing in the framework addresses capital rationing.	With a R3 to R5 billion pipeline and constrained funding, the question is not whether a project has a positive NPV but whether it is the best use of the next rand. Recommend ranking competing projects by profitability index rather than by NPV.
No hurdle rate is stated, and no one is identified as responsible for setting it.	Without a stated minimum acceptable return, "should the investment proceed" has no benchmark. Recommend a board-approved hurdle rate, differentiated by project risk.
Strategic fit is not an explicit gate, despite the Axion 2030 priorities.	A project can pass the financial tests and still not serve GROW, CONNECT, POWER or MATURE. Recommend an explicit strategic screen before the financial work begins, so capital is not spent evaluating projects that do not fit.
There is no post-investment review.	The framework ends at approval. Without a lookback comparing actual

	outcomes to the business case, forecasting bias is never corrected and the same optimistic assumptions recur. Recommend a post-implementation review for projects above a threshold, reported to the board.
A single-point NPV does not value flexibility.	A 3 to 5 year PPA preserves the option to change course as AI-driven demand becomes clearer; a 25 year asset does not. Where forecasts are this uncertain, that optionality has real value and should at least be described, even if it is not formally valued.
Digital tool outputs are to be independently validated.	Stated on page 8 and a likely task hook. See section 7.